Functions - Maxima, Minima and Monotonicity - Part 2

Course: Pre-Calculus-I

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Exponential Functions: $f(x) = a^x$

Base	Monotonicity	Extrema on $\mathbb{R}$
$a > 1$	increasing on $\mathbb{R}$	none (range $(0, \infty)$)
$0 < a < 1$	decreasing on $\mathbb{R}$	none (range $(0, \infty)$)

Why? For $a > 1$ and $x_2 > x_1$:

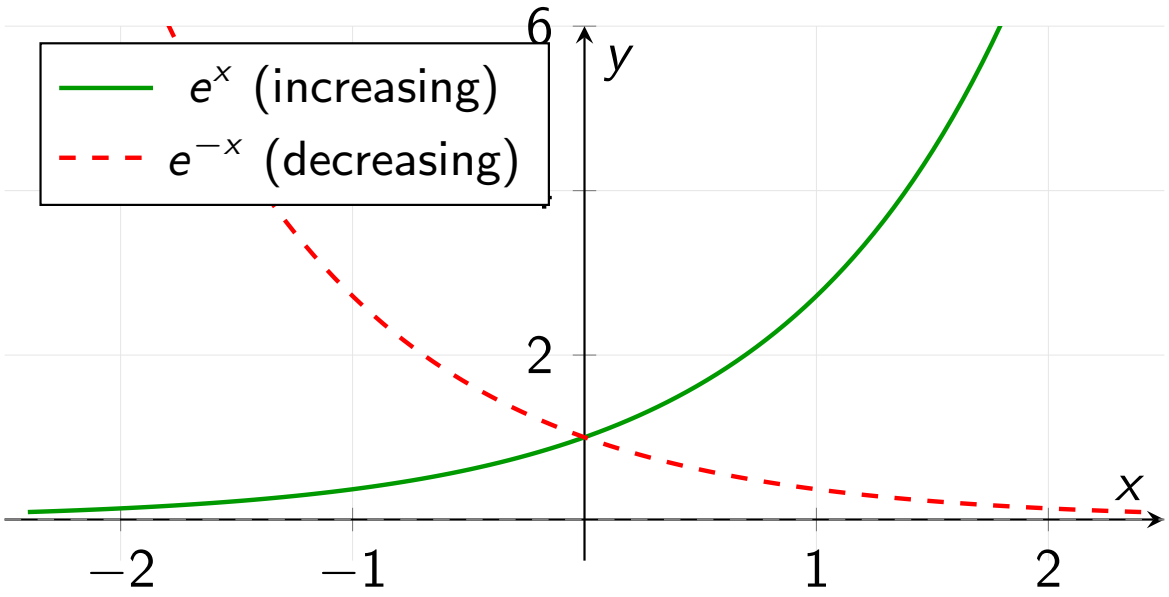
$$a^{x_2} = a^{x_1} \cdot a^{x_2 - x_1} > a^{x_1} \cdot 1 = a^{x_1}.$$

For $0 < a < 1$, the factor $a^{x_2 - x_1} < 1$ reverses the inequality.

Example: $f(x) = e^{-x} = (1/e)^x$ with base $1/e < 1$.

- Decreasing on $\mathbb{R}$; range $(0, \infty)$; no maximum or minimum on $\mathbb{R}$.
- On $[0, 2]$: max at $x = 0$ ($f = 1$), min at $x = 2$ ($f = e^{-2} \approx 0.135$).

Exponential: Growth vs. Decay



Both have horizontal asymptote $y = 0$ and *no* local extrema.

Absolute Value Function: $f(x) = |x|$

Piecewise form:

$$f(x) = |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

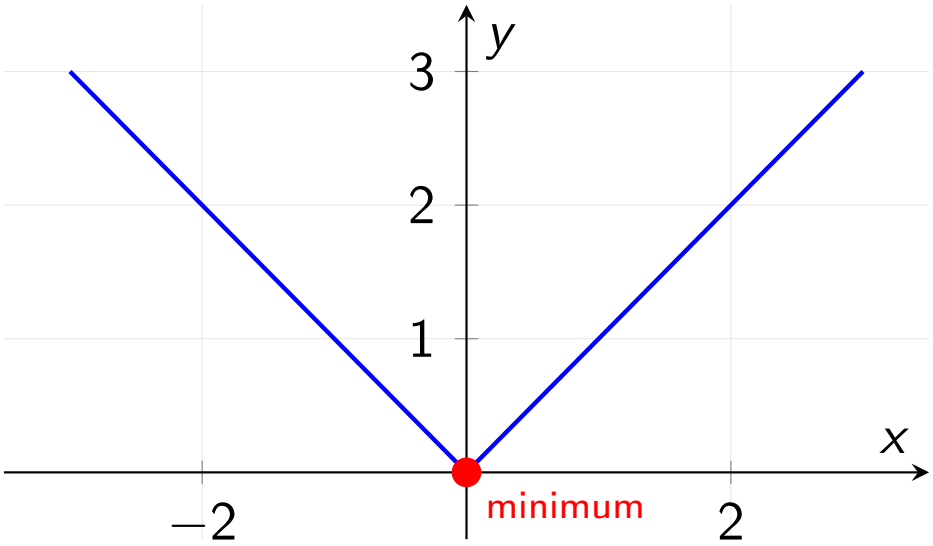
Key facts:

- $|x| \geq 0$ always, and $|0| = 0 \Rightarrow$ absolute minimum at $x = 0$, value 0.
- For $x < 0$: $f(x) = -x$ is decreasing; for $x > 0$: $f(x) = x$ is increasing.
- No maximum on $\mathbb{R}$; range $[0, \infty)$.
- On $[-2, 3]$: max at $x = 3$ ($f = 3$), min at $x = 0$ ($f = 0$).

Note

$|x|$ *measures distance from* 0*; the smallest distance is* 0.

Absolute Value: Graph and Extrema



Transformed example: $g(x) = -2|x-1| + 4$.

- Since $|x-1| \geq 0$, we have $-2|x-1| \leq 0$, so $g(x) \leq 4$.
- Absolute maximum $g(1) = 4$; no minimum on $\mathbb{R}$.
- Increasing on $(-\infty, 1)$, decreasing on $(1, \infty)$.

Trigonometric Functions: $\sin x$ and $\cos x$

Function	Range	Max value	Min value
$\sin x$	$[-1, 1]$	1 (at $x = \frac{\pi}{2} + 2k\pi$)	-1 (at $x = -\frac{\pi}{2} + 2k\pi$)
$\cos x$	$[-1, 1]$	1 (at $x = 2k\pi$)	-1 (at $x = \pi + 2k\pi$)

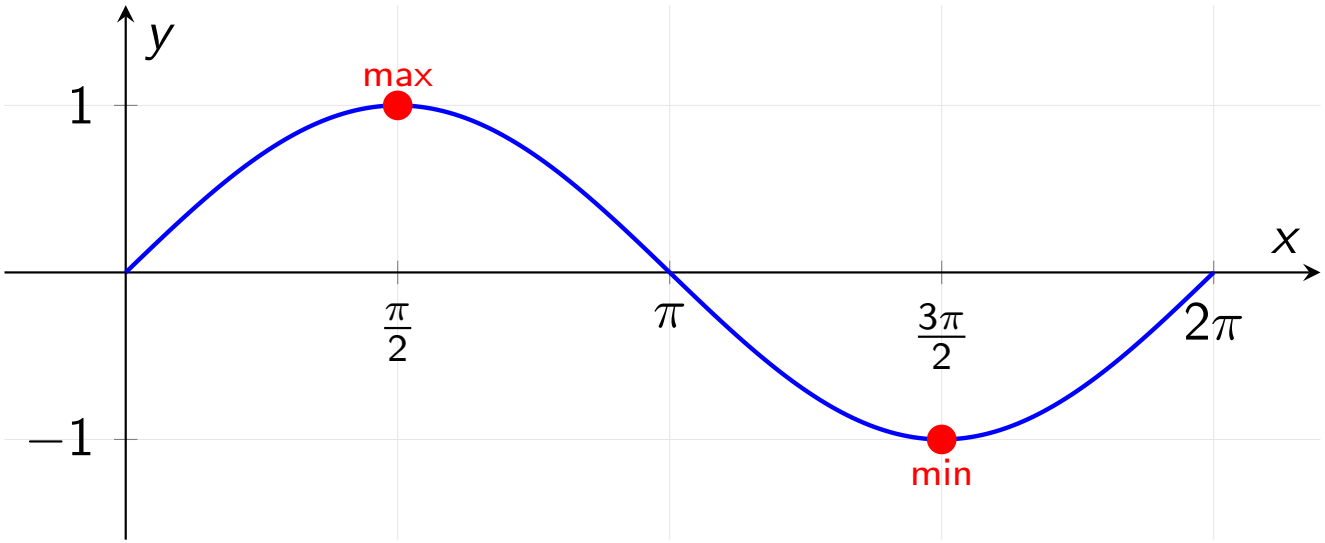
Monotonicity on one period $[0, 2\pi]$:

- $\sin x$: increasing on $[0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi]$; decreasing on $[\frac{\pi}{2}, \frac{3\pi}{2}]$.
- $\cos x$: decreasing on $[0, \pi]$; increasing on $[\pi, 2\pi]$.

Note

These follow directly from the unit circle: the y-coordinate of a point on the circle is $\sin x$ *and the x-coordinate is* $\cos x$.

Trigonometric: Graph of $\sin x$



Example: $f(x) = 3\sin(2x) + 1$

Amplitude: $|A| = 3$, **Vertical shift:** $+1$.

Since $-1 \leq \sin(2x) \leq 1$:

$$-3 \leq 3\sin(2x) \leq 3 \quad \Longrightarrow \quad -2 \leq 3\sin(2x) + 1 \leq 4.$$

Range: $[-2,\ 4]$. Absolute maximum $= 4$, absolute minimum $= -2$.

x (first period)	$f(x)$	Type
$\frac{\pi}{4}$	$3(1) + 1 = 4$	absolute maximum
$\frac{3\pi}{4}$	$3(-1) + 1 = -2$	absolute minimum

$\sin(2x)$ rises then falls then rises again, so f alternates increasing / decreasing.

Worked Example: $f(x) = x^3 - 3x + 1$ (Table Method)

x	-2	-1	0	1	2	2.5	3
$f(x)$	-1	3	1	-1	3	9.125	19

Reading the table:

- From $x = -2$ to $x = -1$: f rises $-1 \to 3 \Rightarrow$ increasing.
- From $x = -1$ to $x = 1$: f falls $3 \to 1 \to -1 \Rightarrow$ decreasing.
- From $x = 1$ onward: f rises $-1 \to 3 \to 9.1 \to 19 \Rightarrow$ increasing.

Local extrema:

- Local maximum $f(-1) = 3$ (peak).
- Local minimum $f(1) = -1$ (valley).
- On $[-2, 3]$: absolute maximum $f(3) = 19$, absolute minimum $f(1) = -1$ (tied with $f(-2) = -1$).

Example: Closed Interval $[a, b]$

Find the absolute extrema of $f(x) = x^3 - 3x + 1$ on $[-2, 3]$ (graph-assisted).

Candidates: interior peak/valley points $x = -1, 1$; endpoints $x = -2, 3$.

x	$f(x)$	Reason
-2	-1	endpoint
-1	3	local peak
1	-1	local valley
3	19	endpoint

Conclusion:

- Absolute maximum: $f(3) = 19$ (at endpoint).
- Absolute minimum: $f(1) = -1$ (at valley; also tied at $x = -2$).

Summary by Function Type

Function	Range	Monotonicity	Extrema
$mx + c$	$\mathbb{R}$	incr. if $m > 0$, decr. if $m < 0$	none (unless restri
$ax^2 + bx + c$	$[k, \infty)$ or $(-\infty, k]$	changes at vertex	vertex: min if a max if $a < 0$
a^x	$(0, \infty)$	incr. if $a > 1$, decr. if $0 < a < 1$	none
$\lvert x \rvert$	$[0, \infty)$	decr. then incr.	min at 0
$\sin x$, $\cos x$	$[-1, 1]$	oscillates	max 1, min -1

General method (no derivatives):

1. Locate peaks and valleys from the graph, table or algebraic form.
2. Use range/bounds to identify absolute extremes.
3. On a closed interval, compare with endpoint values.
4. State monotonicity by reading the graph left to right.

Practice Problems

1. Using a table of values, determine the intervals of increase/decrease and the local extrema of $f(x) = x^3 - 6x^2 + 9x$.
2. Find the absolute extrema of $f(x) = x^2 - 4x + 3$ on $[0, 5]$ (complete the square first).
3. Show that $f(x) = e^{-2x}$ is decreasing on $\mathbb{R}$ by comparing $f(x_1)$ and $f(x_2)$ for $x_1 < x_2$; state its range.
4. For $f(x) = |x - 2| + 1$, find the absolute minimum and the intervals of monotonicity using the piecewise form.
5. Find the maximum and minimum values of $f(x) = 2\sin x + 3$ on $[0, 2\pi]$.
6. The profit function $P(x) = -2x^2 + 40x - 150$ (in thousands) is given for $x \geq 0$. Complete the square to find the level x that maximises profit and the maximum profit.
7. Sketch the graph of $f(x) = x^4 - 2x^2$ by tabulating values and identify all peaks and valleys.

Exercise: Chapter 3, Topic 3.1 — selected questions on extrema and monotonicity.

Key Takeaways

- **Monotonicity** is read from the graph (left to right) or by comparing function values:
  - $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$: increasing;
  - $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$: decreasing.
- **Peaks and valleys** of the graph are local maxima and minima.
- **Quadratics:** completing the square gives the vertex $\Rightarrow$ global extremum.
- **Standard ranges:** $a^x > 0$, $|x| \geq 0$, $\sin x, \cos x \in [-1, 1]$ directly bound max/min values.
- On a **closed interval**, always compare interior peaks/valleys with endpoint values.
- Each function family has its own *range* and *natural extrema*:
  - Linear: none; Quadratic: vertex; Exponential: none;
  - Absolute value: corner minimum; Trigonometric: bounded oscillating extrema.